



Assessment Cover Sheet 2025/2026

Course Code: UC2SAT102	Course Title: Statistical Analysis Tools & Techniques	Lecturer: Isah Lawal
Assignment No: 2	Total number of pages in this document: 4	Maximum Word Count: NA
Assignment Title: Statistical Data Analysis Course Project II		
Date Set: 20.04.2026	Submission Date: 11:59 Sunday, 03.05.2026	Feedback Date: Within 3 weeks afterward

Section 1: Submission

Record of Submission and Plagiarism Declaration

In submitting completed work for assessment, you declare that this submission is your work and that any collaborative work or reuse of existing material has been explicitly indicated in the text. When submitting completed work, you agree that it may be submitted for plagiarism testing.

Work should be submitted by the above deadline:

- Electronically Via Moodle in the first case. If problems are encountered, then via email to the course staff.
- Ensure that you read and follow all instructions for this assignment and any additional instructions provided by the course staff. Failure to do so may adversely affect your grade.

IT IS YOUR RESPONSIBILITY TO KEEP A COPY OF ALL SUBMITTED WORK.

Section 2: Data Analysis and Predictive Modelling

Introduction:

Education is vital for long-term economic growth, and analyzing educational data provides insights into critical questions such as the factors influencing student achievement, the ability to predict academic performance, and the identification of students who may need to repeat a class. Understanding student performance through data analysis and modeling is essential for both educators and students, as it enables the implementation of strategies that will enhance academic outcomes.

In this project, you will analyze real-world data from secondary schools in Portugal to investigate the impact of various factors on students' success. This analysis will encompass school-related factors (such as absenteeism and school choice), demographic factors (including student age and parental education and occupation), and social factors (like time spent with friends and alcohol consumption). The objective is to predict student achievement and identify the key variables that influence success or failure in the educational process.

Dataset:

The dataset for this assessment focuses on students' performance in two subjects: Mathematics and Portuguese. It includes 30 explanatory variables that provide insight into the factors influencing students' success in these areas. A comprehensive description of each variable, along with the dataset itself, can be found on the course page on Moodle for your reference.

Description of Tasks:

Task A:

Conduct a comprehensive statistical analysis of the datasets titled “Students_Performance_Maths” and “Students_Performance_Portuguese” to extract meaningful insights that address the following key questions:

1. How do average grades in Math and Portuguese differ between students who attended nursery school and those who did not, considering variables such as parental education and family size?
2. Is there a statistically significant difference in average Math grades between students who receive paid extra classes and those who do not?
3. Develop distinct profiles of students based on demographic factors, family background, and social characteristics that correlate with significant differences in average grades between Math and Portuguese.

Task B:

- Use the dataset “Students_Performance_Maths” to train a reliable regression model that predicts students' maths scores, adhering to best practices for effective generalization. Ensure that all steps of good modelling practices are followed.
- After training the model, evaluate its performance using appropriate regression metrics. Additionally, derive and document the regression equation for the finalized model.
- Utilize the trained model to estimate maths scores for the 10 students listed in the "Prediction_Sample" file. Present these predicted scores in a table formatted to match the order of the records in the original file.

Task C:

- For the dataset “Students_Performance_Portuguese,” create a new column labelled “Proficiency.” This column should categorize each student’s performance as "Good" or "Not Good" based on whether their scores are above or below the average score of the dataset, respectively.
- Train a classification model to predict students’ proficiency in Portuguese, distinguishing between "Good" and "Not Good."
- Evaluate the classification model's performance using appropriate metrics and report the findings comprehensively.

Report your work:

The report aims to demonstrate your proficiency at interpreting and presenting the results of your data analysis. Use the LaTeX template provided to report your insights and model performances. Emphasis should be placed on professionalism and clarity in the report.

Deliverables:

Submit a single PDF file containing your report using the submission link provided on the course page in Moodle. Submit also the code used in your analyses using the same submission link.

Grading:

The general breakdown of the grade is as follows:

A: Outstanding effort. The student shows exceptional understanding and knows how to choose and apply concepts, can put the analysis into a cogent argument, can identify weaknesses in the analysis, and can communicate results.

B: Good effort. The student shows understanding of most of the concepts, puts together an adequate argument, identifies some weaknesses of their analysis, and communicates most results clearly to others.

C: Passing effort. The student has some understanding of concepts but has some trouble putting analysis together in a cogent argument, and the communication of results is somewhat unclear.

D: Struggling effort. The student is making some effort but has some misunderstanding of some of the concepts and is unable to put together a cogent argument. Communication of results is sometimes unclear.

E: Student is not making a sufficient effort. Demonstrates marginal understanding of the concepts and is unable to choose appropriate techniques for a given task. The communication of the results is unclear.

Section 4: Marking and Assessment

This assignment will be marked out of:	A-F
Contribution to the final course mark:	60%
Estimated number of hours to be spent on this assignment:	18hr